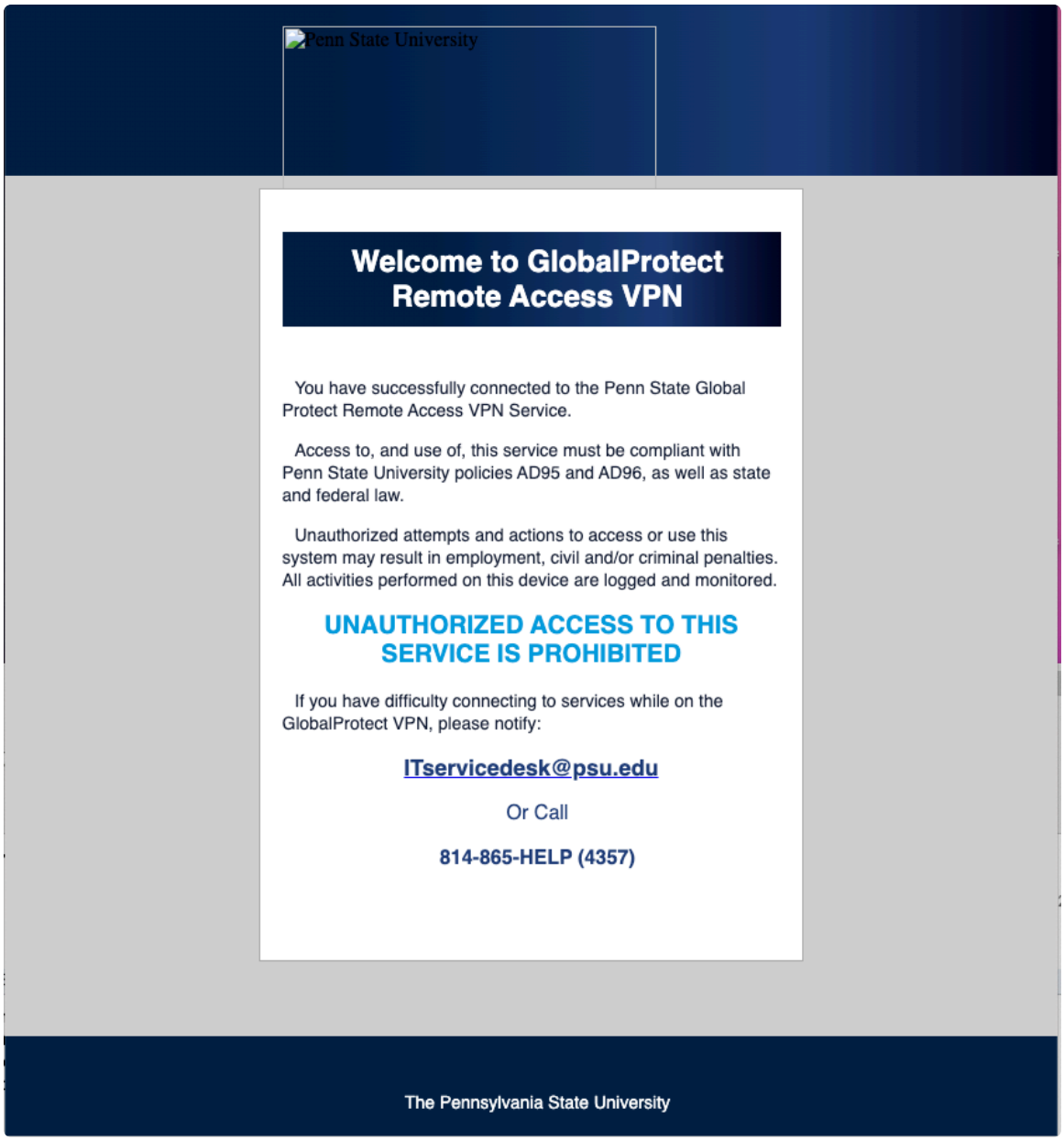
Big Data Lab Cluster User Guide

Important: Please ensure each Part is successful before moving on to the next.

Part 1: Use PSU VPN

1. Access campus network. To access the campus network from off-campus, you must use the PSU VPN (**GlobalProtect**). Once successful, you should see the VPN connected status in your browser.



2. Verify your connection by opening your Terminal (Mac/Linux) or Command Prompt (Windows) and running the following command to ping the cluster gateway:

```
ping 146.186.87.215
```

Press `Ctrl + C` to stop the ping. If you receive continuous responses from that IP address, your connection is active and working.

Part 2: Access the server directly (BDLLINUX1)

Our cluster consists of **4 machines** working together. Your username is your PSU Access ID (the part of your email before the `@` symbol). For example, if your email is `abc1234@psu.edu`, your username is `abc1234`.

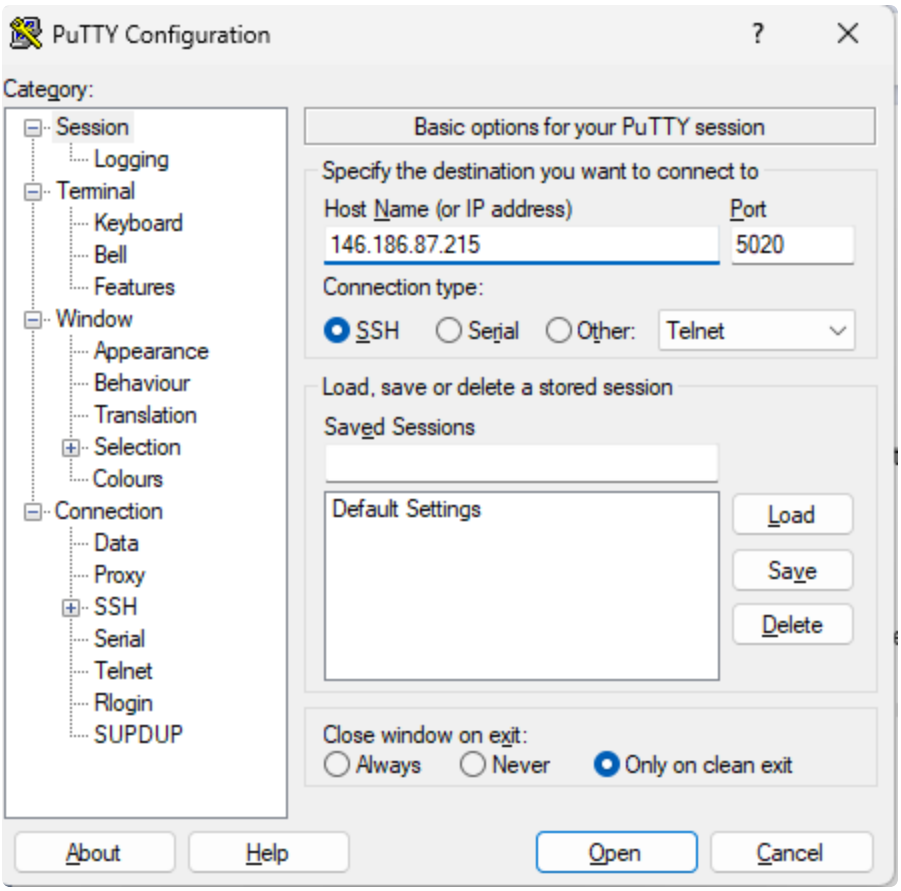
Using Terminal / Command Prompt (Recommended)

Run the following command to connect to the primary machine **BDLLINUX1**

```
ssh abc1234@146.186.87.215 -p 5020
```

Using a GUI Client in Windows (Optional)

If you prefer a graphical interface on Windows, you can use popular applications like **PuTTY**. Please configure your connection exactly as shown in the screenshot below.



Password Details:

- The initial password is: `Bigdatalab%%%%`
- When typing your password in the terminal, no characters or dots will appear on the screen. This is a standard Linux security feature. Just type it blindly and press Enter.

Directly Accessing All Machines:
If you want to bypass the first machine and log in to all 4 machines directly from the outside, you can use the -p option with port number (listed below) so the gateway (146.186.87.215) will forward you to the machine you want to visit.

- BDLLINUX1: 5020
- BDLLINUX2: 5021
- BDLLINUX3: 5022
- BDLLINUX4: 5023

Part 3: Access Servers by Internal Network

Alternatively, you can navigate between the cluster nodes internally. Once you successfully log in to **BDLLINUX1**, you can SSH into the other machines using their internal private IP addresses.

- BDLLINUX1: `192.168.108.95`
- BDLLINUX2: `192.168.108.96`
- BDLLINUX3: `192.168.108.97`
- BDLLINUX4: `192.168.108.95`

For example, to hop from **BDLLINUX1** to **BDLLINUX2**, run:

```
ssh abc1234@192.168.108.96
```

Part 4: Copy Data Between Machines

Assume you want to pull data from a **source server** directly into your **destination server**, and the destination server is your **current** server, you can use the rsync command in your terminal:

```
rsync -avh --partial --info=progress2 -e ssh <username>@<source_server_ip>:
<path_to_source_folder> ~/<path_to_destination_folder>
```

Let us first breakdown the meaning of options in this command

- `-a` (Archive): Preserves file attributes. It copies everything recursively while keeping the original modification times, permissions, owners, and symlinks intact.

- `-v` (Verbose): Increases verbosity. The terminal will explicitly list every single filename as it gets copied so you can monitor the active transfer.
- `-h` (Human-readable): Formats numbers into a readable format. It outputs file sizes and transfer speeds in KB, MB, or GB instead of long strings of bytes.
- `--partial` (Resume Support): Prevents data loss during network drops. If the connection fails mid-transfer, it keeps the partially downloaded files. Running the command again will safely resume from where it left off.
- `--info=progress2` (Global Progress Bar): Displays a clean, single-line progress indicator. It updates the overall percentage, current speed, and Estimated Time of Arrival (ETA) without spamming your terminal screen.
- `-e ssh` (Secure Shell): Instructs `rsync` to encrypt the network traffic using the SSH protocol, ensuring your data remains private during transit between the two servers. Sometimes, if you can not access the source sever from your current destination server, and
  need port forwarding (assuming 5001 is the port number), then use `-e "ssh -p 5001"` instead.

All variables enclosed in angled brackets `< >` are placeholders. You must replace them with your own details before running the command.

- `<username>` : Your login account name on the **source server**.
- `<source_server_ip>` : The IP address of the **source server** hosting the data.
- `<path_to_source_folder>` : The absolute directory path on the **source server** where the data currently sits.
- `<path_to_destination_folder>` : The directory path inside your home directory (`~`) on your **current (destination) server** where you want to store the data.

Part 5: Familiarize Yourself with Common Terminal Commands

Since all communication with the cluster happens over SSH, most of your work will be done via the command line. You should familiarize yourself with the following essential tools:

- **slurm**: The cluster workload manager used for submitting and scheduling computation jobs.
- **scp / rsync**: Command-line utilities for transferring files and data between your local machine and the cluster.
- **nano / vim**: Built-in text editors for modifying files directly inside the terminal.
- **apt**: Using `apt` to install system-wide software

Using `apt` to install system-wide software requires `sudo` (administrator) privileges. Regular users do not have this. If your project strictly requires a system-level package, please contact

me at cbw5803@psu.edu. Most software installations do *not* require admin privileges. For example, you can install Python packages locally using `pip3 install --user <package_name>`